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Developmental Biology

TMC7 deficiency causes acrosome biogenesis defects and male infertility in mice

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Abstract

Transmembrane channel-like (TMC) proteins are a highly conserved ion channel family consisting of eight members (TMC1–TMC8) in mammals. TMC1/2 are components of the mechanotransduction channel in hair cells, and mutations of TMC1/2 cause deafness in humans and mice. However, the physiological roles of other TMC proteins remain largely unknown. Here, we show that *Tmc7* is specifically expressed in the testis and that it is required for acrosome biogenesis during spermatogenesis. $Tmc7^{-/-}$ mice exhibited abnormal sperm head, disorganized mitochondrial sheaths, and reduced number of elongating spermatids, similar to human oligo-astheno-teratozoospermia. We further demonstrate that TMC7 is colocalized with GM130 at the cis-Golgi region in round spermatids. TMC7 deficiency leads to aberrant Golgi morphology and impaired fusion of Golgi-derived vesicles to the developing acrosome. Moreover, upon loss of TMC7 intracellular ion homeostasis is impaired and ROS levels are increased, which in turn causes Golgi and endoplasmic reticulum (ER) stress. Taken together, these results suggest that TMC7 is required to maintain pH and ion homeostasis, which is needed for acrosome biogenesis. Our findings unveil a novel role for TMC7 in acrosome biogenesis during spermiogenesis.

eLife assessment

This study reports an **important** discovery highlighting the essential role of the putative ion channel, TMC7, in acrosome formation during sperm development and thus male fertility. The evidence for the requirement of TMC7 in acrosome biogenesis and sperm function is **convincing**, although its function as an ion channel remains to be further determined. Overall, this work will be of great interest to developmental biologists and ion channel physiologist alike.

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Introduction

Spermiogenesis is the post-meiotic phase of male germ cell development during which the haploid spermatids differentiate into mature spermatozoa[1]. Mouse spermiogenesis consists of 16 steps characterized by morphological changes in acrosome formation, nuclear condensation and elongation, flagellum formation, and cytoplasm removal[2]. Acrosome biogenesis is one of the key events of spermiogenesis, and it begins with the initial stage of spermatid development. The acrosome is a specialized organelle with a cap-like structure covering the anterior part of the mature spermatozoon head and is indispensable for the fertilization process[3]. Although acrosome biogenesis is thought to be derived from the Golgi complex in early spermatids, growing evidence suggests that the endoplasmic reticulum (ER) and lysosomes are involved in acrosome biogenesis[4]. Defects in acrosome biogenesis can result in subfertility or infertility.

Acrosome biogenesis is generally classified into four successive phases, namely the Golgi, cap, acrosome, and maturation phases[5]. A number of ER and Golgi-associated proteins have been shown to be involved in acrosome biogenesis. Heat shock protein 90B1 (HSP90B1) is an ER molecular chaperone that has been shown to be a testis-specific protein, and HSP90B1 deficiency leads to a globozoospermia-syndrome like phenotype[6]. GM130 (Golgi matrix protein 130) localize on the cis surface of the Golgi apparatus, and loss of GM130 causes acrosome malformations and results in male infertility[7]. GOPC (Golgi-associated PDZ and coiled-coil motif-containing) is a trans-Golgi-associated protein that plays a role in vesicle transport from the Golgi apparatus. GOPC deficiency leads to vesicle transport failure, which in turn results in malformation of the acrosome and to infertility[8]. Other vesicle transport proteins including PICK1, ICA1l, and VPS13b are involved in proacrosomal vesicle trafficking during acrosome formation[9,10]. Recent evidence suggests that autophagy is involved in proacrosomal vesicle fusion and transport to form the acrosome, and disruption of ATG7[11,12] or ATG5 causes aberrant acrosome formation and male infertility or subfertility. Despite these findings, further studies are needed to better understand the complexity of acrosome biogenesis.

Transmembrane channel-like (TMC) proteins are highly conserved ion channel-like proteins, with eight family members (TMC1–TMC8) identified in humans and mice [13–16]. TMC proteins are predicted to have 6–10 transmembrane domains, and all protein members contain a conserved 120 amino acid TMC domain[15]. The single-particle cryo-electron microscopy structure shows that TMC1 assembles as dimers and that each monomer contains 10 transmembrane domains with four domains (S4–S7) that line the channel pore[17]. Tmc1 and 2 are mechano-electrical transducer (MET) channels in cochlear hair cells, and mutations in *Tmc1/2* cause deafness in humans and mice[18] by disrupting hair cell mechanoelectrical transduction, and *Tmc* gene therapy can restore auditory function and balance in mice[19,20]. Germline mutations in *Tmc6* and *Tmc8* are associated with epidermodysplasia verruciformis, which leads to increased susceptibility to HPV infection [14,15]. Among the eight TMC protein members, TMC1 and TMC2 are well studied due to their role in hearing conduction, and understanding the physiological and functional roles of other TMC family members remains a major challenge.

In this study, we found that *Tmc7* begins to be expressed in diplotene spermatocytes and that it is expressed at high levels in round spermatids in mouse testes. By generating *Tmc7* conventional knockout mice (*Tmc7*$^{-/-}$), we demonstrate that TMC7 deficiency leads to aberrant Golgi morphology and impaired fusion of Golgi-derived vesicles to the acrosome, thus resulting in male infertility with an oligo-astheno-teratozoospermia (OAT) phenotype. Our findings provide new insights into the physiological role of TMC family members in spermiogenesis.

Results

Tmc7 expression is confined to the testis, and deletion of *Tmc7* leads to male infertility

To explore the roles of *Tmc* family members during spermatogenesis, we analyzed the expression of *Tmc1*–*Tmc8* using the mouse developmental transcriptome (E-MTAB-6798)[21]. Among the eight *Tmc* family members, we found that only *Tmc5* and *Tmc7* exhibited elevated expression in the testis during development (**Fig. 1a**). *Tmc7* caught our great interest because *Tmc7* was expressed at higher levels than *Tmc5* in the testis during development. We next analyzed *Tmc7* expression in different tissues using single-cell transcriptomic data from the Mouse Cell Atlas database[22]. We found that *Tmc7* was predominantly expressed in the testis when compared with all other analyzed tissues (**Fig. 1b**). Further, we analyzed *Tmc7* in different stages of germ cell development using single-cell sequencing data of male-specific tissues and organs in mice and humans[23]. Interestingly, *Tmc7* began to be expressed in diplotene spermatocytes and was expressed at high levels in round spermatids in mouse testes (**Fig. 1c, d**). The *Tmc7* expression pattern was confirmed by RT-PCR (**Fig. 1e, f**), and *Tmc7* was again found to be highly expressed in round spermatids of human testis (Fig. S1a, b). The distinct expression pattern in the testis indicates the potential role of *Tmc7* in spermatogenesis.

We generated *Tmc7*-deficient mice ($Tmc7^{-/-}$) by partial deletion of exon 2 (the first 241 bp) using the CRISPR/Cas9 system (**Fig. 1g**). $Tmc7^{-/-}$ mice were confirmed by PCR-based genotyping and immunoblotting of the separated cytoplasmic and membrane testis proteins (Fig. S1c, d). We then examined the phenotypes and found that $Tmc7^{-/-}$ mice were viable and had a normal appearance and exhibited a normal metabolic rate (Fig. S1e-j). However, $Tmc7^{-/-}$ male mice showed smaller testes (**Fig. 1h, i**) and a reduced testis/body weight ratio (**Fig. 1j**) compared to control mice. Moreover, $Tmc7^{-/-}$ male mice exhibited complete infertility (Table S1). No difference in the number of tubules per testicle or in the number of Sertoli cells was seen when comparing $Tmc7^{-/-}$ and WT mice; Sertoli cell were identified by immunofluorescence staining of the Sertoli cell marker SOX9 (Fig. S1k, i). These findings suggest that TMC7 is essential for spermatogenesis and thus for male fertility.

TMC7 deficiency leads to severe defects in spermiogenesis

To determine the cause of infertility in $Tmc7^{-/-}$ mice, we examined the composition of cell types in seminiferous tubules and cauda epididymis sections using hematoxylin staining in mouse testes at different developmental stages. In seminiferous tubules, the populations of germ cells were comparable between wild type (WT) and $Tmc7^{-/-}$ mice at postnatal day (PD)14 and PD21. Intriguingly, the number of elongating spermatids in $Tmc7^{-/-}$ mice decreased significantly compared with WT mice at PD28 (Fig. S2a). We next examined the testis and cauda epididymis sections of 9-week-old WT and $Tmc7^{-/-}$ mice. We observed spermiogenesis defects in $Tmc7^{-/-}$ mice as evidenced by the reduced number of elongating spermatids, the increased number of sloughed germ cells, and large vacuoles in the seminiferous tubules and cauda epididymis (**Fig. 2a, b**). The vacuolation became more serious with increasing age (Fig. S2a). The total number of sperm in the cauda epididymis was drastically decreased in $Tmc7^{-/-}$ mice (~0.15×10^6 cells) compared to wild type (WT) mice (~8.0×10^6 cells) (**Fig. 2c**). The morphological assessment of spermatozoa from the caudal epididymis revealed that a large proportion of spermatozoa (~70%) in $Tmc7^{-/-}$ mice exhibited abnormal head morphology, similar to human OAT (**Fig. 2d, e**). In parallel with this, periodic acid Schiff (PAS) staining indicated that a large number of spermatids failed to differentiate into elongated spermatids with sickle-shaped heads and instead exhibited abnormal head morphology. The total number of spermatids began to significantly decrease at steps 9–12 of spermiogenesis (**Fig. 2f, g**), and we performed Mito-tracker staining and found shortened or

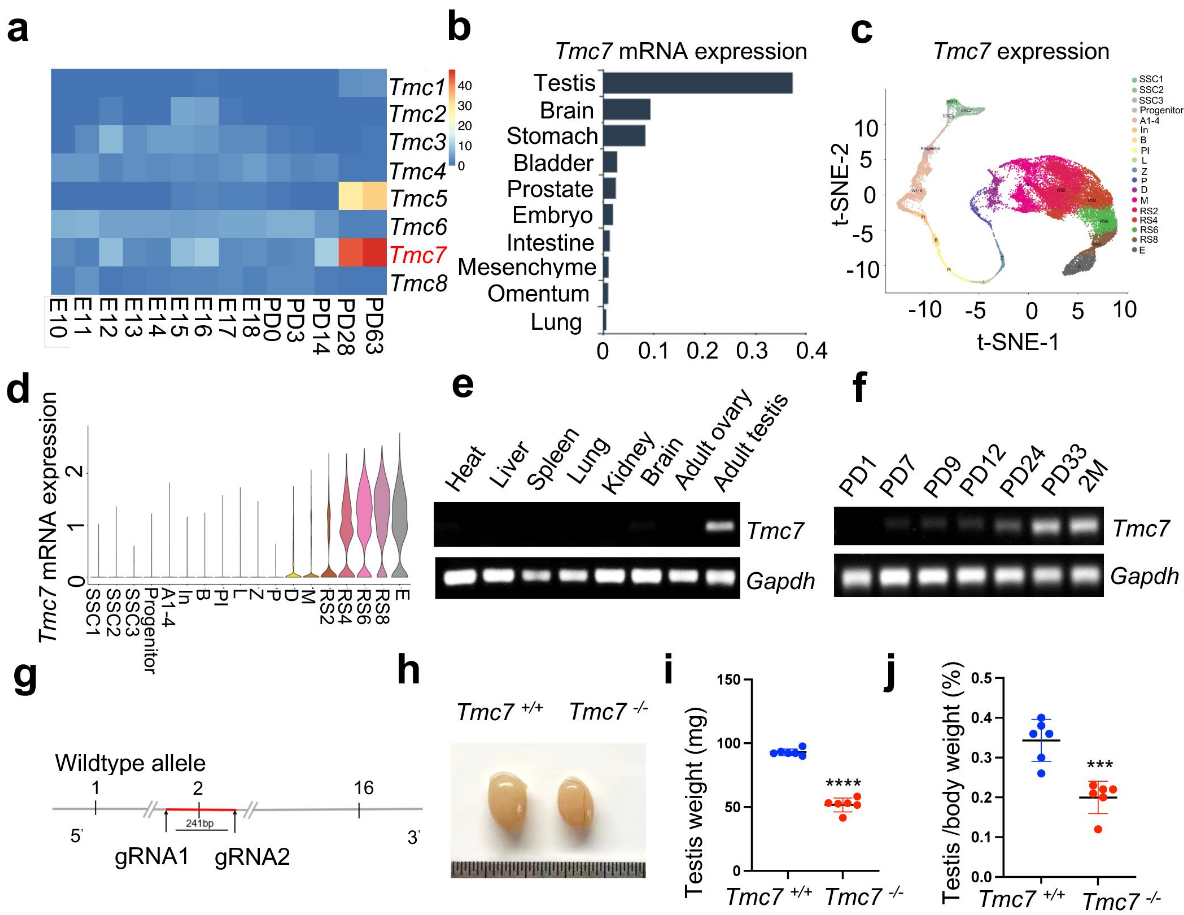

Figure 1.

Tmc7 expression is confined to the testis, and its deletion leads to male infertility.

(a) Heatmap analysis of the expression of *Tmc* family members at embryonic day (E)10–E18 and different postnatal days (P0, P3, PD14, PD28, and PD63) in mouse testis using the Evo-devo mammalian organ database (*http://apps.Kaessmannlab.org/evodevoapp/*). (b) The analysis of *Tmc7* expression in different tissues using the Mouse Cell Atlas database (*https://bis.zju.edu.cn/MCA/*). (c, d) T-Distributed Stochastic Neighbor Embedding (t-SNE) plot and violin plot showing the expression of *Tmc7* in different mouse germ cells using the Male Health Atlas (*http://malehealthatlas.cn/*). SSC1-3: Spermatogonial stem cells I-III; Progenitor: spermatogonia progenitor cell; A1–4: A1–4 spermatogonia; In: Intermediate spermatogonia; B: Type B spermatogonia; PI: Preleptotene; L: Leptotene; Z: Zygotene; P: pachytene; D: Diplotene; M: Metaphase; RS: Round spermatid. E: Elongating spermatid. (e) The expression of *Tmc7* in different adult mouse tissues was measured by RT-PCR, including the heart, liver, spleen, lung, kidney, brain, ovary, and testis. *Gapdh* was used as the control. (f) The expression of *Tmc7* at different postnatal times (PD1, PD7, PD9, PD12, PD24, PD33, and 2 months) was measured by RT-PCR. *Gapdh* was used as the control. (g) Schematic illustrating the generation of *Tmc7*$^{-/-}$ mice using CRISPR/Cas9. (h) Testis from 9-week-old WT and *Tmc7*$^{-/-}$ mice. (i) Testicular weights of 9-week-old WT and *Tmc7*$^{-/-}$ mice. Six adult mice of each genotype were used. (j) Testis/body weight ratio of 9-week-old WT and *Tmc7*$^{-/-}$ mice. Six mice of each genotype were used. Images are representative of at least three independent experiments. The results are presented as the mean ± S.D., and two-sided unpaired Student's t-tests were used to calculate the P-values (*P < 0.05, **P < 0.01, ***P < 0.001).

disorganized mitochondrial sheaths in *Tmc7*$^{-/-}$ spermatozoa collected from the cauda epididymis (**Fig. 2h**). Meanwhile, no motile or progressive sperm were found in the *Tmc7*$^{-/-}$ mice (**Fig. 2i, j**). Moreover, meiotic progression was not affected in *Tmc7*$^{-/-}$ mice as evidenced by immunofluorescence staining of the meiosis markers SYCP3 (red) and γH2AX (green) in the seminiferous tubules of PD20 WT and *Tmc7*$^{-/-}$ mice (Fig. S3a). Overall, we concluded that TMC7 ablation results in defects in spermiogenesis and in OAT-like phenotypes.

Acrosome biogenesis is impaired starting from the Golgi phase in *Tmc7*$^{-/-}$ spermatids

Acrosome biogenesis is one of the key events of spermiogenesis. To investigate whether TMC7 is required for the proper formation of the acrosome, FITC-labelled peanut agglutinin (PNA) was used to assess the acrosomal status of spermatids in 9-week-old WT and *Tmc7*$^{-/-}$ mice. The spermatids of WT mice had normal structures in the Golgi, cap, acrosome, and maturation phases, while spermatids in *Tmc7*$^{-/-}$ mice exhibited abnormal acrosome morphology as early as the Golgi phase. Fragmented and malformed acrosomes were observed in all four phases of spermatids in the *Tmc7*$^{-/-}$ mice (**Fig. 3a**). Absent or fragmented acrosomes were also found in *Tmc7*$^{-/-}$ spermatids collected from the cauda epididymis (Fig. S4a). To better understand the morphologic defects of the acrosome in greater detail, transmission electron microscopy (TEM) was performed to analyze the acrosome structure in WT and *Tmc7*$^{-/-}$ mice. We found that large vesicles accumulated adjacent to developing acrosomes in all phases of acrosome biogenesis, starting from the Golgi phase (**Fig. 3b**, left panel) in the spermatids of *Tmc7*$^{-/-}$ mice. Additionally, abnormal acrosome structures and defective nuclear condensation were observed in *Tmc7*$^{-/-}$ spermatids (**Fig. 3b**, right panels). Further investigation revealed that knockout of *Tmc7* resulted in Golgi apparatus fragmentation and disorganization, as determined by TEM and immunofluorescence staining of the Golgi marker GM130 (**Fig. 3b-c**). The aspect ratio (height over width) of the Golgi apparatuses in *Tmc7*$^{-/-}$spermatids (R = ~2.9) based on TEM images were also significantly smaller than in WT spermatids (R = ~4.9) (**Fig. 3d**). Meanwhile, we found that the Golgi body failed to correctly orient toward the acrosome (Fig. S4b). We further performed immunoblotting to measure Golgi stress markers, HSP47 and TFE3 in 3 and 9-week-old *Tmc7*$^{-/-}$ testes compared with WT testes. The results showed that the protein levels of HSP47 and TFE3 were significantly increased upon loss of TMC7, indicating the occurrence of Golgi stress (**Fig. 3e**, Fig. S4c). GOPC is localized in the trans-Golgi network and is involved in the transport and fusion of the proacrosomal vesicles with the acrosome[8]. By immunostaining for GOPC and PNA, we found that GOPC staining (red) was colocalized with PNA (green) in WT spermatids. In contrast, colocalized GOPC and PNA spermatids were significantly decreased in *Tmc7*$^{-/-}$ mice (**Fig. 3f**). Together these findings suggest that TMC7 is required for proacrosomal vesicle trafficking and fusion during acrosome biogenesis.

TMC7 localizes to the cis-Golgi, and this is required for maintaining Golgi integrity

To further delineate the role of TMC7 in acrosome formation, we next sought to determine the subcellular localization of TMC7 in the testis. We performed immunostaining and found that TMC7 was predominantly expressed in the perinuclear regions of pachytene spermatocytes and round spermatids in WT testes but not in *Tmc7*$^{-/-}$ testes, suggesting the specificity of the antibody (**Fig. 4a**). By immunofluorescence co-staining of TMC7 with the cis-Golgi marker GM130, the trans-Golgi marker TGN46, and PNA, we found that TMC7 colocalized closely with GM130 but not with TGN46 (**Fig. 4b, c**, Fig. S5a), indicating its localization to the cis-Golgi apparatus. Further evaluation showed that TMC7 and PNA were closely associated but not colocalized (**Fig. 4d**, Fig. S5b). In parallel, we carried out co-immunoprecipitation with TMC7 antibody coupled with

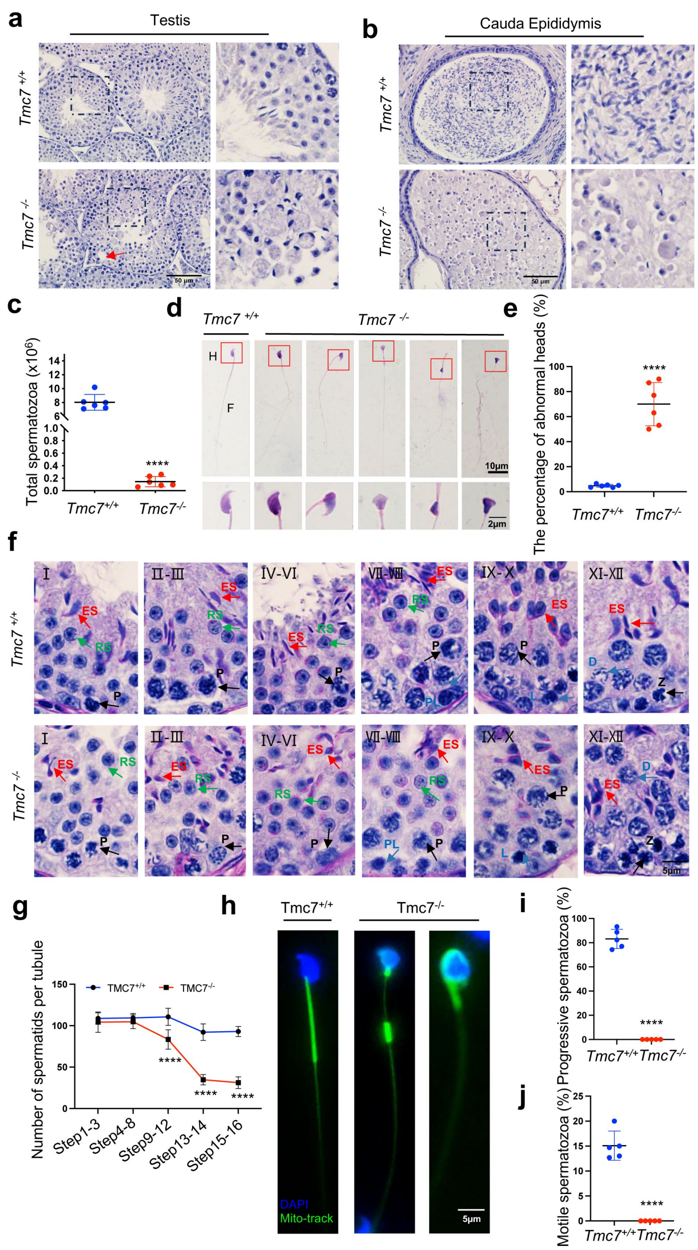

Figure 2.

TMC7 deficiency leads to severe defects in spermiogenesis.

(a, b) Hematoxylin staining in testis and caudal epididymis sections of 9-week-old WT and *Tmc7*$^{-/-}$ male mice. Scale bar, 50 μm. (c) The total number of sperm in the caudal epididymis from 9-week-old WT and *Tmc7*$^{-/-}$ male mice. Six mice were used. (d) Hematoxylin-eosin staining of the spermatozoa isolated from the caudal epididymis in 9-week-old WT and *Tmc7*$^{-/-}$ male mice. H: head, F: flagella. Scale bar, 10 μm, 2 μm. (e) The percentage of spermatozoa with abnormal heads in the caudal epididymis from WT and *Tmc7*$^{-/-}$ male mice. Six mice were used. (f) PAS and hematoxylin staining in 9-week-old WT and *Tmc7*$^{-/-}$ mouse seminiferous tubules. The numbers represent the stages of spermatogenesis. P: pachytene spermatocyte, PL: preleptotene spermatocyte, L: leptotene spermatocyte, Z: zygotene spermatocyte, D: diplotene spermatocyte, RS: round spermatid, ES: elongating spermatid. Scale bar, 5 μm. (g) Quantification of spermatid numbers in seminiferous tubules from spermatid stages 1–16. N = 15 tubules per stage and per genotype; 3 mice per genotype were used and 5 tubules per stage and per mouse were counted. (h) Mitochondrial sheath staining by Mito-tracker in sperm collected from the caudal epididymis. Scale bar, 5 μm. (i, j) The percent of progressive and motile spermatozoa in 9-week-old WT and *Tmc7*$^{-/-}$ mice. Five mice were used. Images are representative of at least three independent experiments. The results are presented as the mean ± S.D., and two-sided unpaired Student's t-tests were used to calculate the P-values. (*P < 0.05, **P < 0.01, ***P < 0.001).

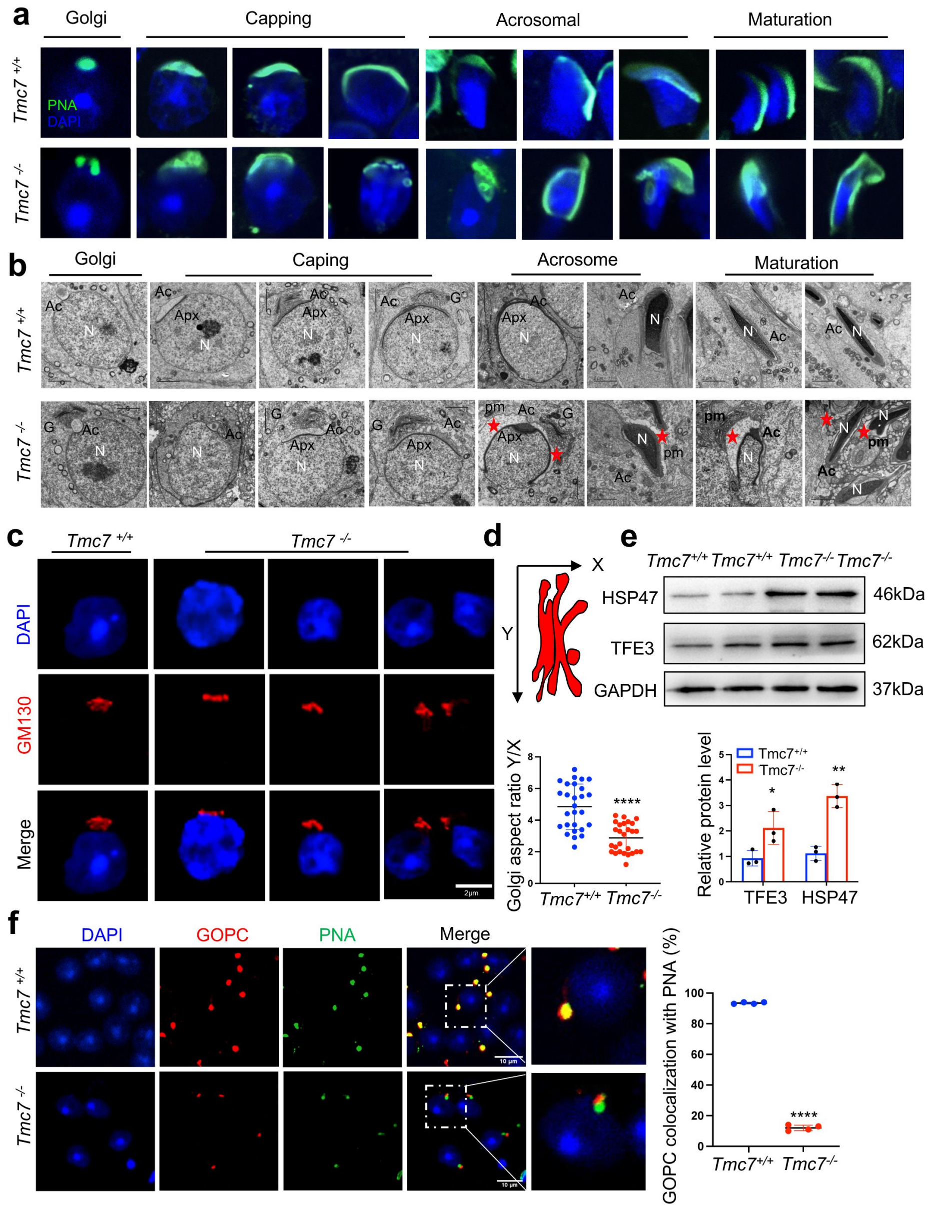

Figure 3.

Acrosome biogenesis is impaired starting from the Golgi phase in *Tmc7*$^{-/-}$ spermatids.

(a) Immunofluorescence staining of PNA (green) in the phases of acrosome biogenesis in 9-week-old WT and *Tmc7*$^{-/-}$spermatids. Nuclei were stained with DAPI (blue). The phases of acrosome biogenesis and corresponding spermiogenesis steps are the Golgi phase (steps 1–3), cap phase (steps 4–7), acrosome phase (steps 8–12), and maturation phase (steps 13–16). (b) TEM images of testicular sections of 9-week-old WT and *Tmc7*$^{-/-}$ mice. The red star represents the abnormal plasma membrane space and accumulated vesicles. N: nucleus, Ac: acrosome; Apx: acroplaxome; pm: plasma membrane. G: Golgi complex; Scale bar, 2 µm. (c) Immunofluorescence staining with the antibody against GM130 in spermatids from the testes of 9-week-old WT and *Tmc7*$^{-/-}$ mice. Scale bar, 2 µm. (d) Calculation of the aspect ratio (height [X] over width [Y]) of 9-week-old WT and *Tmc7*$^{-/-}$ Golgi apparatuses based on TEM images. N = 27, and 9 tubules in each testis were counted from 3 mice. (e) Western blot analysis of protein levels of Golgi stress-associated proteins (HSP47, TFE3) in 3-week-old WT and *Tmc7*$^{-/-}$ testes. Image J was used to quantify the protein level. (f) Immunofluorescence staining of GOPC (red) and PNA (green) and the quantitative analysis of GOPC and PNA lectin colocalization in squashed tubules of 9-week-old WT and *Tmc7*$^{-/-}$ mice. Nuclei were stained with DAPI (blue). Scale bar, 10 µm. Four mice were used, and 200 cells were counted per mouse. Images are representative of at least three independent experiments. The results are presented as the mean ± S.D., and two-sided unpaired Student's t-tests were used to calculate the P-values. (*P < 0.05, **P < 0.01, ***P < 0.001).

immunoblotting in mouse testes and found that TMC7 was able to pull down GM130, the GM130-interacting protein P115, and GRASP65 (**Fig. 4e**), further confirming that TMC7 is localized to the cis-Golgi apparatus.

GM130 is reported to be involved in maintaining the function and integrity of the Golgi apparatus[7]. To gain more insight into the role of TMC7 in the Golgi apparatus, we performed immunostaining against GM130, P115, and GRASP65 in the testes of 9-week-old WT and *Tmc7*$^{-/-}$ mice. The results showed that in the same stage seminiferous tubules, the protein levels of GM130, P115, and GRASP65 were significantly reduced in *Tmc7*$^{-/-}$ testes compared to WT testes (**Fig. 4f, g**, Fig. S5c-e). Immunoblotting further verified the reduced levels of GM130, P115, and GRASP65 upon loss of TMC7 in the testes of 3 and 9-week-old mice (**Fig. 4h**, Fig. S5f). These results suggest that TMC7 colocalizes to GM30 in the cis-Golgi and that this is required for maintaining Golgi integrity.

TMC7 depletion impairs cellular homeostasis and leads to spermatid apoptosis

Intracellular ion homeostasis is essential for proper function, including membrane trafficking, protein glycosylation, and protein sorting[24]. The Golgi apparatus has been shown to be another intracellular Ca^{2+} store in addition to the endoplasmic reticulum (ER)[25]. *Tmc1* mutation leads to reduced Ca^{2+} permeability, thus triggering hair cell apoptosis and subsequent deafness[26]. We measured intracellular Ca^{2+} by flow cytometry in WT and *Tmc7*$^{-/-}$ germ cells isolated from PD21 and PD30 testes. The results showed that the intracellular Ca^{2+} concentration was significantly reduced in *Tmc7*$^{-/-}$ germ cells compared to WT germ cells (**Fig. 5a**, Fig. S6a). In contrast, intracellular pH was increased in *Tmc7*$^{-/-}$ germ cells compared to WT germ cells (**Fig. 5b**, Fig. S6b). These results suggest that the pH and ion homeostasis was impaired. We further measured the reactive oxygen species (ROS) levels and observed that ROS levels were significantly increased upon loss of TMC7 (**Fig. 5c**, Fig. S6c) In addition, we examined ER stress-related proteins by immunoblotting and found that p-EIF2α, CHOP, and GRP78 were significantly increased in 3- and 9-week-old *Tmc7*$^{-/-}$ testes compared with WT testes, indicating the occurrence of ER stress (**Fig. 5d**, Fig. S6d). Moreover, we carried out TUNEL staining in the seminiferous tubules and cauda epididymis. There was no difference in the number of TUNEL-positive cells between 3-week-old WT and *Tmc7*$^{-/-}$ mice of seminiferous tubules. The TUNEL-positive cells in seminiferous tubules and cauda epididymis were significantly increased in 9-week-old *Tmc7*$^{-/-}$ mice compare to WT mice (**Fig. 5e**). The expression of apoptosis-related proteins was next measured by immunoblotting in 9-week-old WT and *Tmc7*$^{-/-}$ mice, depletion of TMC7 increased the levels of Cleaved-CASPASE-3 and BAX and decreased the levels of BCL2 (**Fig. 5f**). These results support the hypothesis that TMC7 is required to maintain intracellular pH and ion homeostasis and that this is needed for acrosome biogenesis.

Discussion

Among the eight TMC protein members, TMC1 and TMC2 are localized to the site of MET channels at the tips of the shorter stereocilia of cochlear hair cells[27,28], and mutations in *Tmc1* and *Tmc2* cause deafness. However, TMC1 and TMC2 expressed in heterologous cell lines remain in the ER and fail to localize to the plasma membrane[18]. TMC4 is a voltage-dependent chloride channel that is predominantly expressed in the taste buds in the posterior region of the tongue[29]. TMC6 and TMC8 are localized to the ER and play crucial roles in zinc transport[30]. However, the physiological function of TMC7 has not been defined. In this study, we demonstrated that *Tmc7* was predominantly expressed in testes (**Fig. 1e**), and by generating knockout mice lacking TMC7 we found that TMC7 deficiency resulted in complete male infertility with an OAT-like phenotype. *Tmc7*$^{-/-}$ mice showed various defects, such as abnormal sperm head, disorganized mitochondrial

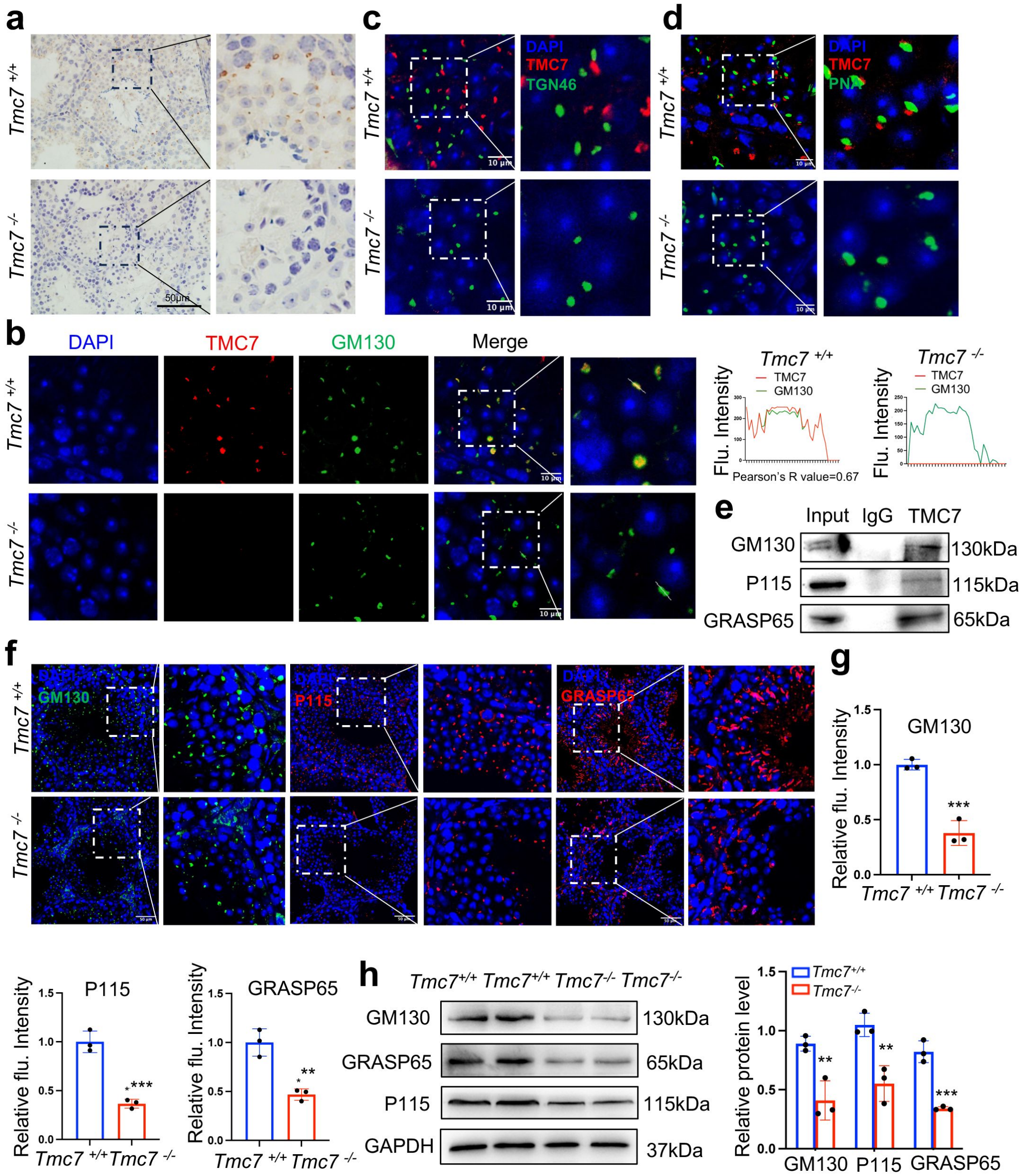

Figure 4.

TMC7 localizes to the cis-Golgi, and this is required for maintaining Golgi integrity.

(a) Immunohistochemical staining with an antibody against TMC7 in seminiferous tubules of 9-week-old mouse testes. Scale bar, 50 μm. (b) Immunofluorescence staining of TMC7 (red) and GM130 (green, a marker of the cis-Golgi), with DAPI indicating the nuclei in seminiferous tubules of 9-week-old WT and *Tmc7*$^{-/-}$ testis. Scale bar, 10 μm. Co-localization was quantified by Image J, and 20 cells were analyzed for the Pearson's R-value. (c) Immunofluorescence staining of TMC7 (red) and TGN46 (green, a marker of the trans-Golgi), with DAPI indicating the nuclei in the seminiferous tubule. Scale bar, 10 μm. (d) Immunofluorescence staining of TMC7 (red) and PNA (green, a marker of the acrosome), with DAPI indicating the nuclei in the seminiferous tubule. Scale bar, 10 μm. (e) Western blot was used to analyze the results of co-immunoprecipitation and showed the interaction between TMC7 and GM130, P115, and GRASP65 in 9-week-old WT testes. (f, g) Immunofluorescence staining and quantification of GM130 (green), P115 (red) and GRASP65 (red) in testicular sections (L-L stage) of 9-week-old WT and *Tmc7*$^{-/-}$ mice. Scale bar, 50 μm. (h) Western blots of the protein levels of GM130 and the GM130-interacting proteins P115 and GRASP65 in 3-week-old WT and *Tmc7*$^{-/-}$ testes. Image J was used to quantify the protein level. Images are representative of at least three independent experiments. The results are presented as the mean ± S.D., and two-sided unpaired Student's t-tests were used to calculate the P-values. (*P < 0.05, **P < 0.01, ***P < 0.001).

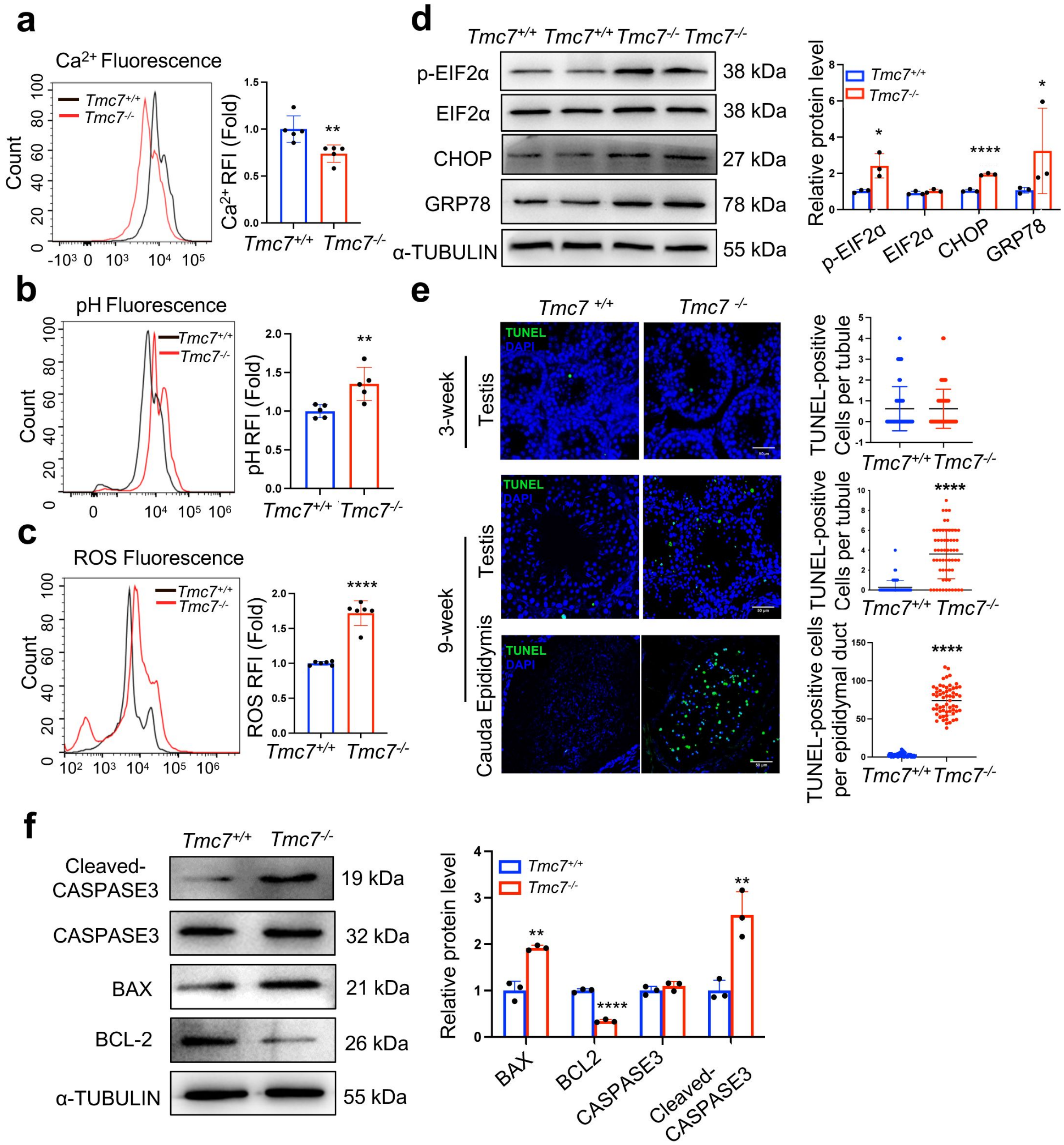

Figure 5.

TMC7 depletion impairs cellular homeostasis and leads to spermatid apoptosis.

(a) The Ca^{2+} levels were measured by flow cytometry in WT and *Tmc7*$^{-/-}$ germ cells isolated from 3-week-old testes. Five mice were used. (b) The pH levels were measured by flow cytometry in WT and *Tmc7*$^{-/-}$ germ cells isolated from 3-week-old testes. Five mice were used. (c) The ROS levels were measured by flow cytometry in WT and *Tmc7*$^{-/-}$ germ cells isolated from 3-week-old testes. Six mice were used. (d) Western blot analysis of protein levels of ER chaperone-associated proteins (p-EIF2α, EIF2α, CHOP, and GRP78) in 3-week-old WT and *Tmc7*$^{-/-}$ testes. Image J was used to quantify the protein level. (e) TUNEL assay and the number of TUNEL-positive cells per tubule in seminiferous tubules and cauda epididymis of 3 and 9-week-old WT and *Tmc7*$^{-/-}$ male mice. Scale bar, 50 μm. N = 60, 20 tubules in each testis and cauda epididymis were counted, and three mice were used. (f) Apoptosis-related proteins (Cleaved-CASPASE3, CASPASE3, BAX, BCL2) were detected by western blotting in 9-week-old WT and *Tmc7*$^{-/-}$ testes. Image J was used to quantify the protein level. Images are representative of at least three independent experiments. The results are presented as the mean ± S.D., and two-sided unpaired Student's t-tests were used to calculate the P-values. (*P < 0.05, **P < 0.01, ***P < 0.001).

sheaths, and reduced number of elongating spermatids. Interestingly, we found that TMC7 co-localized with GM130, but not with TGN46 (**Fig. 4b, c**), as illustrated by the immunofluorescence co-staining assay, indicating that TMC7 located on the cis side of the Golgi apparatus.

Our results demonstrate the critical role of TMC7 during the formation of the acrosome. The acrosome is thought to be derived from the Golgi apparatus, so we speculated that TMC7 deficiency might cause defects in acrosome formation because TMC7 localized on the Golgi apparatus. We observed fragmented and malformed acrosomes and proacrosomal vesicles were found to accumulate around the acrosomes (**Fig. 3b**). Similarly, TMC1 deficiency leads to membrane blebbing and PS externalization at the apical region of hair cells. TMC1 is not only an ion channel but also a lipid scramblase, which is required to maintain homeostasis in hair cells[31]. TMC7 has similar amino acid sequence and predicted structure with TMC1/2 (Fig. S7a, b), suggesting TMC7 might function as a lipid scramblase and an ion channel. Further investigations are required to verify the potential functions of TMC7 as a lipid scramblase and an ion channel in spermatid.

In accordance with our findings, other proteins that localize on the Golgi apparatus have also been reported to be involved in acrosome biogenesis. GM130 deficiency results in acrosome biogenesis failure and male infertility because proacrosomal vesicles are unable to fuse to form acrosomes[30]. Our findings showed that TMC7 was co-localized with GM130 and that TMC7 depletion significantly reduced the protein level of GM130. GOPC is localized on the trans-Golgi region, and its deficiency leads to defective fusion of Golgi-derived transport vesicles to the acrosomal cap and to a complete lack of acrosomes[8]. Colocalized GOPC and PNA spermatids were significantly decreased in *Tmc7*$^{-/-}$ spermatids compared to WT spermatids. Further, GOLGA3, PICK1, and several other Golgi proteins are also required for acrosome biogenesis[32]. Our work, combined with previous studies, broadens our understanding of role the Golgi apparatus plays in the formation of acrosomes.

Our findings highlight the role of TMC7 in maintaining cellular homeostasis. Upon TMC7 depletion, the Golgi apparatus undergoes significant disorganization and is unable to correctly orient toward the acrosome. The mechanism behind this is that intracellular Ca^{2+} concentrations decreased while the pH value was increased in *Tmc7*$^{-/-}$ germ cells compared to WT germ cells, suggesting that TMC7 is essential for maintaining the intracellular pH and ion homeostasis. Our findings further revealed that TMC7 deficiency led to increased ROS levels, Golgi and ER stress, ultimately resulting in apoptosis of spermatids. The Na^+/H^+ exchanger NHE8 is involved in the maintenance of intracellular pH and ion homeostasis, and NHE8 deficiency leads to Golgi-derived vesicles failing to form large acrosomal granules and the acrosomal cap[33]. SLC9A3 is another Na^+/H^+ exchanger, and it is expressed in the acrosomal region of spermatids and loss of SLC9A3 causes severe acrosomal defects[34]. These studies, together with our current findings, support the hypothesis that TMC7 is essential for maintaining the intracellular pH and ion homeostasis, which is needed for the proper progression of acrosome biogenesis.

In conclusion, our findings show that TMC7 localizes to the cis-Golgi region in spermatids and that this is essential for acrosome biogenesis and for male fertility. In the absence of TMC7, intracellular pH and ion homeostasis is disrupted, and this leads to Golgi disorganization and failure to correctly orientate to the acrosome. Subsequently, Golgi-derived proacrosomal vesicles are unable to fuse with the developing acrosome and acrosome biogenesis is severely impaired, and thus TMC7-deficent male mice are completely infertile. Our study highlights the physiological role of TMC family members in spermatogenesis.

Materials and methods

Generation of *Tmc7* knockout mice

Tmc7 conventional knockout mice in the C57BL/6J background were obtained from Cyagen Bioscience (China). Exon 2 was selected as the target site, and Cas9 mRNA and single guide RNA were co-injected into fertilized eggs for knockout mouse production. The pups were genotyped by PCR using the following primers: F1:5'-AGA TGG CTT AGT GGG TGC CTT AGT-3'; R1:5'TGA TGG ACA GAG AGA TGA TGA ATG G-3'; F3: 5'-CAG TCC CAG GTT CAG TTC AGT GAG-3'. In WT mice, the primer pair amplified a 513 bp PCR product, while in *Tmc7*$^{-/-}$ mice it amplified an 891 bp product. Mice were housed under a controlled illumination regime (light for 12 h) at 21°C–22°C and 60%L±L10% relative humidity. Food and water were freely available, and all animal experiments were conducted in accordance with the guidelines of the Animal Care and Use Committee at Shandong University (ECSBMSSDU2022-2-113).

Metabolic assessment in mice

A Columbus Instruments Comprehensive Lab Animal Monitoring System CLAMS6 was used for the experiment. Nine-week-old WT and *Tmc7*$^{-/-}$ male mice (n = 4) were weighed and had the metabolic rates measured over a period of 48 hours. The rate of oxygen consumption, RER (breathing entropy), energy expenditure, and food intake were automatically recorded every eight minutes. These data were analyzed with a 12 h dark:12 h light cycle at 22–23 °C.

RNA isolation and PCR

The tissues were homogenized in TRIzol (Invitrogen, 15596-026) with a grinder. Following centrifugation, the RNA in the supernatant was obtained by precipitating with isopropanol and washing with ethanol. A total of 1 μg RNA was reverse-transcribed into cDNA with HiScript III RT SuperMix for RT-PCR (+gDNA wiper) (Vazyme, R223-01) according to the manufacturer's instructions. The cDNA was used to detect the expression of Tmc7 with Taq Master Mix (Vazyme, P112-03) with the following primers: F – AGC CAA TCC ATC CGC AAG TAT; R – TGC TGC TG TAC CGT TTC CAC. The products were analyzed by agarose gel electrophoresis.

Sperm collection

The cauda epididymis was collected and placed in an EP tube containing PBS. Sperm were released for 30 min in an atmosphere of 5% CO_2 at 37 °C. The suspension was diluted at 1:50 and transferred to a hemocytometer for counting. For the anther experiments, the PBS with sperm was spread on the slide and fixed with 4% paraformaldehyde (Servicebio, G1101). The slides were stored at –80°C after drying.

Sperm motility assessment

For each genotype, at least three mice were used for the sperm motility assessment. 10μL sperm suspension was added to the 80 mm glass cell chamber (Hamilton Thorne, 80μm). More than 200 sperm were analyzed through Animal Motility system (Hamilton Thorne, Beverly, MA, USA), the analysis included the total motility, average path velocity, straight-line velocity, and linear velocity. Each analysis was replicated at least five times.

Histological analysis

For the morphology analysis of the sperm from the caudal epididymis, sperm smears were directly stained with hematoxylin and sealed with neutral balsam. The tissue was fixed in 4% paraformaldehyde overnight, then dehydrated and embedded in paraffin. Paraffin-embedded tissues were sectioned at 4 μm. Tissue slides were deparaffinized in xylene and dehydrated in a series of ethanol. The tissues were then stained by hematoxylin or with periodic acid staining using a PAS Stain Kit (Abcam, ab150680) according to the manufacturer's instructions. Finally, tissue slides were rehydrated in a series of ethanol and xylene and sealed with neutral balsam. An Olympus BX53 microscope system was used for imaging.

Immunofluorescence

We performed immunofluorescence of germ cells in the testes using two methods, including testis slides and squashed seminiferous tubules. The squashed seminiferous tubules were prepared according to a previous report[35]. The testis slides were deparaffinized in xylene and rehydrated in a series of ethanol. Antigen retrieval was performed by heating in EDTA Antigen Retrieval Solution (pH 9.0) (ZSGB-BIO, ZLI-9096), and nonspecific binding sites were blocked with EZ-Buffers M 1X Block BSA in PBS (Sangon, C520035-0250). The slides were incubated with primary antibody overnight at 4L. Primary antibodies included anti-TMC7 (1:200 dilution, Abcam, ab191521), anti-SOX9 (1:500 dilution, Sigma, AB5535), anti-GOPC (1:200 dilution, Proteintech, 12163-1-AP), anti-SYCP3 (1:500 dilution, Santa Cruz, sc-33195), anti-H2AX (1:500 dilution, Millipore, 05-636-1), anti-GM130 (1:100 dilution, 610822, BD Biosciences), anti-P115 (1:200 dilution, Proteintech, 13509-1-AP), and anti-GRASP65 (1:200 dilution, Proteintech, 10747-2-AP). The next day they were incubated with the secondary antibody (1:500 dilution, Thermo, A-11070 and A-11012) for 1 h at room temperature. The Alexa Fluor 488 conjugates of lectin PNA (1:500 dilution, Thermo, L21409), anti-GM130 (1:200 dilution, Proteintech, CL488-11308), and anti-TGN46 (1:200 dilution, Proteintech, CL488-13573) were used as secondary antibodies. For the staining of Mito-track (1:5000 dilution, Beyotime, C1048), the sperm released from the caudal epididymis were directly stained for 30 min at 37 °C. The sperm were then spread on the slide and observed. A TUNEL assay was performed using a one-step TUNEL Apoptosis Assay Kit (KeyGEN BioTECH, KGA7072) according to the manufacturer's instructions. Briefly, the slide was deparaffinized and antigen retrieval was performed. dUTP was used to label the cells, and the nuclei were stained with DAPI (Abcam, ab285390). Fluorescent images were captured on a Dragonfly 200 confocal microscopy system (Andor Technology).

Western blotting

The membrane and cytoplasmic proteins of the testis were obtained using a membrane protein extraction kit (Abmart, A1008) according to the manufacturer's instructions. Total protein in the testis was obtained by grinding in a tissue grinder in RIPA lysis buffer (Beyotime, P0013B), and the concentration of protein was measured using a BCA protein assay kit (Beyotime, P0009). The proteins were separated by SDS-PAGE and then transferred to a PVDF membrane. The membranes were blocked in 5 % non-fat milk and then incubated with primary antibodies overnight at 4 °C. Primary antibodies included anti-TMC7 (Dia-an Biotech), anti-ATP1A1 (Proteintech, 14418-1-AP), anti-GAPDH (Proteintech, 60004-1-1g), anti-Cleaved-CASPASE3 (Cell Signaling Technology, 9661), anti-CASPASE3 (Cell Signaling Technology, 9662), anti-BAX (Cell Signaling Technology, 41162), anti-BCL2 (Cell Signaling Technology, 17447), anti-α-TUBULIN (Proteintech,11224-1-AP), anti-P-EIF2α (ABclonal, AP0745), anti-EIF2α (ABclonal, A0764), anti-GRP78 (Proteintech, 11587-1-AP), anti-CHOP (Proteintech, 15204-1-AP), anti-P115 (Proteintech, 13509-1-AP), and anti-GRASP65 (Proteintech, 10747-2-AP). All antibodies were diluted 1:1000. The next day, the membrane was incubated with horseradish peroxidase-conjugated secondary antibody for 1 hour, and the protein was detected using an ECL system (Vazyme, E412-02). The quantitation was performed in Image J. Additional replicates of the western blots have been included in revised Fig. S8.

Chromosome spreading

Meiotic nuclear spreading was performed as previously reported[36]. Briefly, spermatocytes released from seminiferous tubules were incubated in hypotonic solution for 1 hour and then transferred to 100 mM sucrose. The suspension was spread on the glass slides with 1% PFA and

0.15% Triton X100, and dried slides were stored at –80°C or were used for the immunofluorescence staining.

Transmission electron microscopy

The testes of 9-week-old WT and *Tmc7*$^{-/-}$ mice (n = 3) were used for the TEM analysis. The testes were washed in PBS three times and then fixed in 1% O_SO_4 in 0.1 M PBS (pH 7.4) buffer at 4°C overnight. The testes were then fixed in 1% O_SO_4 at room temperature for 1 hour. The samples were then washed with water three times for 15 min each and dehydrated in a graded ethanol series (50%, 70%, 95%, and 100%). The samples were finally embedded with resin, cut into 80 nm sections, and viewed with a JEM-1400 transmission electron microscope.

Flow cytometry

The testes of PD21and PD30 WT and *Tmc7*$^{-/-}$ mice (n = 3-6) were used to obtain germ cells. The tunica albuginea was removed and placed into PBS with 120 U/ml Type I Collagenase (Gibco, 17100-017) on a shaker at 37°C and 220 rpm for 10 min. The supernatant was filtered through a 70 μm cell filter screen into a new tube. The sediment was digested again with 0.25% trypsin-EDTA (Gibco, 25200–072) on a shaker at 37°C and 220 rpm for 10 min and then neutralized by serum albumin and filtered. The germ cells were obtained by centrifuging twice at 1000 × *g* for 5 min and then removing the supernatant.

The levels of Ca^{2+} (Beyotime, Fluo-3 AM, S1056), pH (Beyotime, BCECF AM, S1006), and ROS (Beyotime, DCFH-DA, S0033S) were measured using fluorescent probes according to the manufacturer's protocols. The probes were all diluted 1:2000. Germ cells were stained with the probes at 37°C for 30 min, then washed with PBS three times. The supernatant was filtered through a 45 μm cell filter screen and analyzed by flow cytometry (Gallios, US).

Co-immunoprecipitation

Testes of 9-week-old WT mice were ground in a tissue grinder in RIPA Lysis Buffer (Beyotime, P0013B) to obtain the total protein. TMC7 (Dia-an Biotech) and IgG (Beyotime, A7016) antibodies were incubated with the proteins overnight. The next day, they were incubated with protein A/G magnetic beads (MCE, HY-K0202) for 2 hours. The beads were wash and boiled for 10 min in 1× SDS loading buffer. Western blotting was used for the analysis of input and immunoprecipitated samples with anti-P115 and anti-GRASP65 antibodies.

Statistical analysis

GraphPad Prism 9 was used to analyze the data, and the results are shown as the mean ± standard deviation (S.D.). Statistical significance was determined using two-sided Student's t-tests or two-way analysis of variance (ANOVA). *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$ indicate the level of statistical significance, and ns stands for not significant.

Author Contributions

J.W., Y.Y, performed most experiments, and data analysis, and writing the manuscript. Z.W., L.Y., performed public database sequence data analysis. C.Q., Y.G., L.Y., helped with animal and cell experiment. F.G., G.Lu., Z.C., X.Z., is involved technical, material support and guided the experiment. H.L., Z.L. Supervised the project and revised the manuscript. Zhaojian Liu was the lead contact for the study. Final approval: All authors.

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Data availability

All data generated or analyzed in this study are provided in the article and supplemental information.

Conflict of interest

The authors declare no competing interests.

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Editors

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Senior Editor
Kathryn Cheah
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Reviewer #1 (Public Review):

Summary:

TMC7 knockout mice were generated by the authors and the phenotype was analyzed. They found that Tmc7 is localized to Golgi and is needed for acrosome biogenesis.

Strengths:

The phenotype of infertility is clear, and the results of TMC7 localization and the failed acrosome formation are highly reliable. In this respect, they made a significant discovery regarding spermatogenesis.

In the original version, I pointed out the gap between their pH/calcium imaging data and the hypothesis of ion channel function of TMC7 in the Golgi. Now the author agrees and has changed the description to be reasonable. Additional experiments were also performed, and I can say that they have answered my concern adequately.

I would say it is good to add any presumed mechanism for the observed changes in pH and calcium concentration in the cytoplasm this time.

https://doi.org/10.7554/eLife.95888.2.sa2

Reviewer #2 (Public Review):

Summary:

This study presents a significant finding that enhances our understanding of spermatogenesis. TMC7 belongs to a family of transmembrane channel-like proteins (TMC1-8), primarily known for their role in the ear. Mutations to TMC1/2 are linked to deafness in humans and mice and were originally characterized as auditory mechanosensitive ion channels. However, the function of the other TMC family members remains poorly characterized. In this study, the authors begin to elucidate the function of TMC7 in acrosome biogenesis during spermatogenesis. Through analysis of transcriptomics datasets, they elevated levels of TMC7 in round spermatids in both mouse and human testis. They then generate Tmc7-/- mice and find that male mice exhibit smaller testes and complete infertility. Examination of different developmental stages reveals spermatogenesis defects, including with reduced sperm count, elongated spermatids and large vacuoles. Additionally, abnormal acrosome morphology are observed beginning at the early-stage Golgi phase, indicating TMC7's involvement in proacrosomal vesicle trafficking and fusion. They observed localization of TMC7 in the cis-Golgi and suggest that its presence is required for maintaining Golgi integrity, with Tmc7-/- leading to reduced intracellular Ca2+, elevated pH and increased ROS levels, likely resulting in spermatid apoptosis. Overall, the work delineates a new function of TMC7 in spermatogenesis and the authors propose that that its ion channel and/or scramblase activity is likely important for Golgi homeostasis. This work is of significant interest to the community and is of high quality.

Strengths:

The biggest strength of the paper is the phenotypic characterization of the TMC7-/- mouse model, which has clear acrosome biogenesis/spermatogenesis defects. This is the main claim of the paper and it is supported with the data that are presented.

Weaknesses:

It isn't clear whether TMC7 functions as an ion channel from the current data presented in this paper, but the authors are careful in their interpretation and present this merely as a hypothesis supporting this idea.

https://doi.org/10.7554/eLife.95888.2.sa1

Reviewer #3 (Public Review):

Summary:

In this study, Wang et al. have demonstrated that TMC7, a testis-enriched multipass transmembrane protein, is essential for male reproduction in mice. Tmc7 KO male mice are sterile due to reduced sperm count and abnormal sperm morphology. TMC7 co-localizes with GM130, a cis-Golgi marker, in round spermatids. The absence of TMC7 results in reduced levels of Golgi proteins, elevated abundance of ER stress markers, as well as changes of Ca^{2+} and pH levels in the KO testis. However, further confirmation is required because the analyses were performed with whole testis samples in spite of the differences in the germ cell composition in WT and KO testis. In addition, the causal relationships between the reported anomalies await thorough interrogation

Strengths:

By using PD21 testes, the revised assays have consolidated that depletion of TMC7 leads to a reduced level of Ca^{2+} and an elevated level of ROS in the male germ cells. The immunohistochemistry analyses have clearly indicated the reduced abundance of GM130, P115, and GRASP65 in the knockout testis.

Weaknesses:

The Discussion section contains sentences reiterating the Introduction and Results of this manuscript (e.g., Lines 79-85 and 231-236; Lines 175-179 and 259-263). Those read repetitive and can be removed.

Future studies are required to decipher how TMC7 stabilizes Golgi structure, coordinates vesicle transport, and maintains the germ cell homeostasis.

https://doi.org/10.7554/eLife.95888.2.sa0

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *TMC7 knockout mice were generated by the authors and the phenotype was analyzed. They found that Tmc7 is localized to Golgi and is needed for acrosome biogenesis.*
>
> *Strengths:*
>
> *The phenotype of infertility is clear, and the results of TMC7 localization and the failed acrosome formation are highly reliable. In this respect, they made a significant discovery regarding spermatogenesis.*
>
> *Weaknesses:*
>
> *There are also some concerns, which are mainly related to the molecular function of TMC7 and Figure 5.*
>
> *(1) It is understandable that TMC7 exhibits some channel activity in the Golgi and somehow affects luminal pH or Ca2+, leading to the failure of acrosome formation. On the other hand, since they are conducting the pH and calcium imaging from the cytoplasm, I do not think that the effect of TMC7 channel function in Golgi is detectable with their methods.*

We agree with the reviewer that there are no direct evidences showing the effect of TMC7 channel function in Golgi. We have changed the description in the revised manuscript.

> *(2) Rather, it is more likely that they are detecting apoptotic cells that have no longer normal ion homeostasis.*

We thank the reviewer for raising this concern. We apologize for not labeling the postnatal stage in original Figure 5. We measured intracellular Ca2+, pH and ROS in PD30 testes (revised Fig. S6a-c), no apoptotic cells were observed at this stage (revised Fig. S6e, f). Apoptotic cells were found in the seminiferous tubules and cauda epididymis of 9-week-old *Tmc7–/–* mice (revised Fig. 5e-f). We have included TUNEL data in testis of PD21, PD30 and 9-week-old mice (revised Fig. 5e, f and Fig. S6e, f). In accordance with our findings, *Tmc1* mutation has also been shown to result in reduced Ca2+ permeability, thus triggering hair cell apoptosis (Fettiplace, R, PNAS. 2022) [1].

> *(3) Another concern is that n is only 3 for these imaging experiments.*

As suggested by the reviewer, more replicates were included in imaging experiments.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *This study presents a significant finding that enhances our understanding of spermatogenesis. TMC7 belongs to a family of transmembrane channel-like proteins (TMC1-8), primarily known for their role in the ear. Mutations to TMC1/2 are linked to deafness in humans and mice and were originally characterized as auditory mechanosensitive ion channels. However, the function of the other TMC family members remains poorly characterized. In this study, the authors begin to elucidate the function of TMC7 in acrosome biogenesis during spermatogenesis. Through analysis of transcriptomics datasets, they identify TMC7 as a transmembrane channel-like protein with elevated transcript levels in round spermatids in both mouse and human testis. They*

then generate Tmc7-/- mice and find that male mice exhibit smaller testes and complete infertility. Examination of different developmental stages reveals spermatogenesis defects, including reduced sperm count, elongated spermatids, and large vacuoles. Additionally, abnormal acrosome morphology is observed beginning at the early-stage Golgi phase, indicating TMC7's involvement in proacrosomal vesicle trafficking and fusion. They observed localization of TMC7 in the cis-Golgi and suggest that its presence is required for maintaining Golgi integrity, with Tmc7-/- leading to reduced intracellular Ca2+, elevated pH, and increased ROS levels, likely resulting in spermatid apoptosis. Overall, the work delineates a new function of TMC7 in spermatogenesis and the authors suggest that its ion channel activity is likely important for Golgi homeostasis. This work is of significant interest to the community and is of high quality.*

Strengths:

The biggest strength of the paper is the phenotypic characterization of the TMC7-/- mouse model, which has clear acrosome biogenesis/spermatogenesis defects. This is the main claim of the paper and it is supported by the data that are presented.

Weaknesses:

The claim is that TMC7 functions as an ion channel. It is reasonable to assume this given what has been previously published on the more well-characterized TMCs (TMC1/2), but the data supporting this is preliminary here, and more needs to be done to solidify this hypothesis. The authors are careful in their interpretation and present this merely as a hypothesis supporting this idea.

We appreciate the insightful comment. It is indeed a limitation of our study that we lack strong evidences to support that TMC7 functions as an ion channel. We have planned to conduct cellular electrophysiology in GC-1 cells heterologous expression of TMC7. However, TMC7 was trapped in the endoplasmic reticulum like TMC1 and TMC2 (Yu X, PNAS. 2020)[2], and failed to localize to the Golgi. According to the reviewer's suggestion, we have made careful and more detailed interpretation the molecular function of TMC7 in the revised manuscript.

Reviewer #3 (Public Review):

Summary:

In this study, Wang et al. have demonstrated that TMC7, a testis-enriched multipass transmembrane protein, is essential for male reproduction in mice. Tmc7 KO male mice are sterile due to reduced sperm count and abnormal sperm morphology. TMC7 co-localizes with GM130, a cis-Golgi marker, in round spermatids. The absence of TMC7 results in reduced levels of Golgi proteins, elevated abundance of ER stress markers, as well as changes of Ca2+ and pH levels in the KO testis. However, further confirmation is required because the analyses were performed with whole testis samples in spite of the differences in the germ cell composition in WT and KO testis. In addition, the causal relationships between the reported anomalies await thorough interrogation.

Strengths:

The microscopic images are of great quality, all figures are properly arranged, and the entire manuscript is very easy to follow.

Weaknesses:

*(1) Tmc7 KO male mice show multiple anomalies in sperm production and morphogenesis, such as reduced sperm count, abnormal sperm head, and deformed

midpiece. Thus, it is confusing that the authors focused solely on impaired acrosome biogenesis.

We are grateful to your comments and suggestions. We agree and have added these defects in spermiogenesis of *Tmc7–/–* mice in the abstract and discussion sections of revised manuscript.

(2) Further investigations are warranted to determine whether the abnormalities reported in this manuscript (e.g., changes in protein, Ca2+, and pH levels) are directly associated with the molecular function of TMC7 or are the byproducts of partially arrested spermiogenesis. Please find additional comments in "Recommendations for the authors".

Thank you for raising this concern. Per your comments, we have included data of intracellular Ca2+, pH and ROS in PD21 testes. The intracellular homeostasis was impaired as early as PD21, indicating TMC7 depletion impairs cellular homeostasis which in turn results in arrested spermiogenesis.

Recommendations for the authors:

Reviewing Editor (Recommendations For The Authors):

As noted by all three reviewers, current flow cytometry data does not necessarily support the 'ion channel' hypothesis, thus the phenotypic analysis is compelling but the molecular mechanism of how TMC7 facilitates acrosome biogenesis remains incomplete. It is highly recommended for the authors to at least discuss or test alternative hypotheses (as reviewer #2 suggested) such as the possibility of acting as 'lipid scramblase'. Also, the authors need to provide further explanation for other morphological defects if TMC7 is truly a functional ion channel in Golgi (and thus later at acrosome), which is also related to the key question of whether TMC7 is a functional ion channel.

We thank the reviewing editor for the comments and suggestions. We agree that our study lack strong evidences to support that TMC7 functions as an ion channel. We have discussed the possibility of TMC7 acting as 'lipid scramblase' as suggested. We have also included data of intracellular Ca2+, pH and ROS in PD21, PD30 testes.

Indeed, *Tmc7–/–* mice exhibits other defects including abnormal head morphology and disorganized mitochondrial sheaths. As TMC7 is localized to the cis-Golgi apparatus and is required for maintaining Golgi integrity. Previous studies on Golgi localized proteins including GOPC (Yao R, PNAS. 2002)[2], HRB (Kang-Decker N. Science. 2001)[3] and PICK1(Xiao N, JCI. 2009)[4] exhibit similar defects in spermiogenesis with *Tmc7–/–* mice. It is possible that defects morphologies in *Tmc7–/–* mice might be due to impaired function of Golgi.

Reviewer #1 (Recommendations For The Authors):

(1) The authors should provide more details about the imaging experiments using FACS. Since they only describe catalog numbers (Beyotime, S1056, S1006, S0033S) for imaging reagents, it is not immediately clear what reagents they actually used. Since they used Fluo3, BCECF, and DCFH, it would be better to mention their names.

Thanks. We have provided more detailed antibody information as suggested.

(2) I am also concerned that in the FACS there is no information at all about laser wavelength and filter properties. This is especially important for BCECF because the

> *wavelength spectrum changes with pH. Also, if there are any positive controls for these imaging reagents, such as ionophores, it would be more convincing to include them.*

Thank you for your comment. Excitation wavelength is 488nm for detecting Ca2+, pH and ROS in FACS. BCECF is the most popular pH probe to monitor cellular pH and the reagent from Beyotime (S1006) has been used by other studies (Chen S, Blood. 2016)[5], (Liu H, Cell Death Dis. 2022)[6]. To make the results more reliable, we have repeated these experiments in PD21 testes (revised Figure 5a-c). No positive controls for these reagents were used in our experiments.

> *(3) As noted above, it is better to avoid directly linking the cell's abnormal ion homeostasis to TMC7 ion channel function in the text. The discussion should be changed to emphasize that the TMC7-deficient cells are apoptotic and that these physiological phenomena are occurring as a side effect of this apoptosis.*

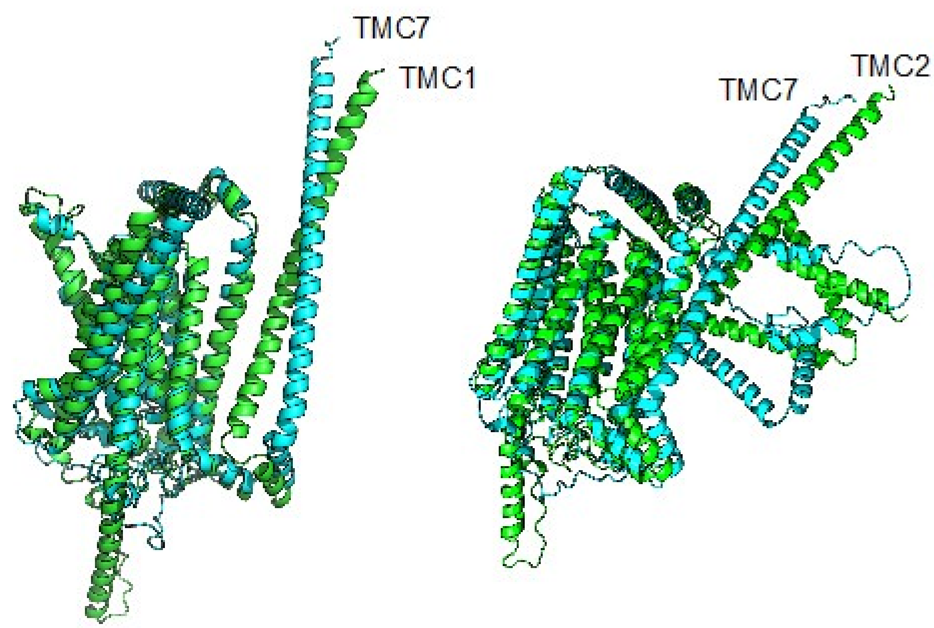
Thank you for raising this concern. We agree with the reviewer that there are no direct evidences showing the effect of TMC7 channel function in Golgi and we have changed the description in the revised manuscript.

We performed new experiment to measure apoptosis and intracellular Ca2+, pH and ROS in PD21 testes. No apoptotic cells were observed at this stage. However, impaired cellular homeostasis was still found in testis of PD21 *Tmc7-/-* mice. These data suggest that TMC7 depletion impairs cellular homeostasis and hence induces spermatid apoptosis.

> *(4) While I understand that it appears to be difficult to experimentally verify the ion channel function of TMC7, it may be supportive to compare its amino acid sequence and/or 3D predicted structure with that of TMC1/2. Including a supplemental figure for this purpose would emphasize the possibility that TMC7 functions as an ion channel.*

We thank the reviewer for making this great suggestion. We compared the amino acid sequence and structure of TMC1, TMC2 with TMC7 respectively. TMC1 had 81% sequence similarity with TMC7 and the RMSD (Root Mean Square Deviation) was 3.079. TMC2 had 82% sequence similarity with TMC7, the RMSD was 2.176. These data suggest that TMC7 has similar amino acid sequence and predicted structure with TMC1/2 and might functions as an ion channel. We have included the predicted structures in revised Fig. S7.

Author response image 1.



> ***Reviewer #2 (Recommendations For The Authors):***
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> *I do not have any experimental comments or concerns to address, but I do ask that the authors consider an alternative hypothesis. Based on prior data demonstrating that TMC1 is a mechanosensitive ion channel, the authors reasonably assume that TMC7 may also function as an ion channel. Although the authors observe alterations in cytosolic Ca2+ and pH upon loss of TMC7 by flow cytometry, which begins to support this hypothesis, these data do not directly demonstrate ion channel activity.*
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> *I was wondering if the authors had considered whether TMC7 could also function as a lipid scramblase. TMC1 has also been proposed to function as a Ca2+-inhibited scramblase, where knockout of TMC1 leads to a loss of phosphatidylserine (PS) exposure and membrane blebbing at the apical region of hair cells (Ballesteros, A. and Swartz, K., Science Advances, 2022). Furthermore, TMC proteins are structurally related to the Anoctamin/TMEM16 family of chloride channels and lipid scramblases, where TMEM16A-B are bona fide Ca2+-activated chloride channels, and TMEM16C-H are characterized as Ca2+-dependent scramblases. Based on their structural similarity and the observation that TMC1 may also exhibit lipid scrambling properties based on the PS exposure, I wonder if the authors may have data that support a TMC7 scramblase hypothesis. I was intrigued by this idea, especially given the authors' observations of large vacuoles in the seminiferous tubules and cauda epididymis and the vesicle accumulation phenotype in their TEM data. Incorporating this hypothesis into the discussion section, at minimum, could provide a valuable perspective, and this line of thought may lead to interesting data interpretation throughout the paper.*

We thank the reviewer for the valuable suggestion. We have discussed the possibility of TMC7 acting as 'lipid scramblase' as suggested.

> ***Reviewer #3 (Recommendations For The Authors):***
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> *(1) Gene symbols should be italicized, and protein symbols should be capitalized.*

Thanks. We have made changes to the manuscript as recommended.

> *(2) Tmc7 KO males show reduced sperm count, which alters the germ cell composition in the testis (Figure 2g). Thus, it is inappropriate to compare protein levels using whole testis lysates (Figure 3e, 4h, 5d, 5f). Instead, the same immunoblotting analyses could be done with purified round spermatids or 3-wk-old testis. Likewise, the significance of the intracellular Ca2+ and pH measurements is potentially diminished by the differences in the germ cell composition in WT and KO mice.*

We appreciate this constructive suggestion. We agree with the reviewer that whole testis lysates diminished the differences between WT and _Tmc7-/-_mice. However, we are unable purify round spermatids due to the lack of specific markers.

> *(3) Figures 2i, 2j: How sperm motility was measured should be specified in the Methods.*

We thank you for your significant reminding and have added sperm motility assessment in Methods section.

> *(4) Figure 4g: It does not make sense to compare the fluorescence intensity of these proteins without making sure that the seminiferous tubules are in the same stage. As shown in Figures S5a and S5b, TMC7 exhibits varied abundance in spermatids at different steps.*

We thank the reviewer for the insightful comment. We have replaced images in the same stage seminiferous tubules and compared the fluorescence intensity of new images as suggested.

> *(5) Figure 4h: How were the band intensities measured? The third band from the left is visually stronger than the first one, but it does not seem to be so according to the column graph. The reviewer measured the intensity of GRASP65 bands relative to alpha-tubulin by ImageJ and obtained relative intensities of 0.35, 0.87, 0.6, and 0.08 for the bands from left to right. Additional replicates of the western blots should be included in the supplementary figures.*

Thank you for this insightful comment. The density and size of the blots were quantified by Image J. We have checked the first band from the left of GRASP65 and it seems that the protein was not fully transferred onto the PVDF membrane. We have performed new experiments and replaced the original bands (Revised Fig. 4h). Additional replicates of the western blots have been included in revised Fig. S8.

> *(6) Figures 5a, 5b: Based on the observation of abnormal intracellular Ca2+ and pH levels in the KO germ cells, the authors concluded that TMC7 maintains the homeostasis of Golgi pH and ion (Lines 223-224, 263-264). However, intracellular Ca2+ and pH levels do not directly reflect those in the Golgi apparatus.*

We thank the reviewer for this important comment. We agree and have changed “Golgi” to “intracellular” as suggested.

> *(7) Figure 5c: ROS is produced during apoptosis. Thus, it is not appropriate to conclude that the increased ROS levels in Tmc7 KO germ cells lead to apoptosis.*

According to the reviewer's comment, we measured ROS and apoptosis in testis of PD21 and PD30 mice. ROS levels were increased, but no apoptotic cells were observed in testis of PD21 and PD30 *Tmc7–/–* mice. Apoptotic cells were observed in testis of 9-week-old *Tmc7–/–* mice (Revised Fig. 5e-f). These data suggest that TMC7 depletion results in the accumulation of ROS, thereby leads to apoptosis.

(1) Fettiplace, R., D.N. Furness, and M. Beurg, The conductance and organization of the TMC1-containing mechanotransducer channel complex in auditory hair cells. Proc Natl Acad Sci U S A, 2022. 119(41): p. e2210849119.

(2) Yu, X., et al., Deafness mutation D572N of TMC1 destabilizes TMC1 expression by disrupting LHFPL5 binding. Proc Natl Acad Sci U S A, 2020. 117(47): p. 29894-29903.

(3) Kang-Decker, N., et al., Lack of acrosome formation in Hrb-deficient mice. Science, 2001. 294(5546): p. 1531-3.

(4) Xiao, N., et al., PICK1 deficiency causes male infertility in mice by disrupting acrosome formation. J Clin Invest, 2009. 119(4): p. 802-12.

(5) Chen, S., et al., Sympathetic stimulation facilitates thrombopoiesis by promoting megakaryocyte adhesion, migration, and proplatelet formation. Blood, 2016. 127(8): p. 1024-35.

(6) Liu, H., et al., PRMT5 critically mediates TMAO-induced inflammatory response in vascular smooth muscle cells. Cell Death Dis, 2022. 13(4): p. 299.

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